

Preface

Fifth International Conference on "**System Modeling & Advancement in Research Trends**" (**SMART-2016**), in technical co-sponsorship with IEEE UP Section, is organised by College of Computing Sciences & Information Technology, Teerthanker Mahaveer University, Moradabad from 25th to 27th November, 2016. This conference brings together leading researchers, engineers and scientists in the domain of interest from around the world by providing a platform to present new advances and research results in the fields of Computational Sciences, system modeling and computer science. The main objective of SMART-2016 is to provide platform for the researchers, engineers and scholars in the field of computer science and information technology to disseminate their latest research results and exchange views on the future research directions in these areas. The purpose of this conference is to make a scientific contribution to the field of computational creativity through discussion and publication on progress in fully autonomous creative systems, modeling human and computational creativity, computational support for human creativity, simulating creativity, and human/machine interaction in creative endeavors. Contributions come from many relevant disciplines, including computer science, artificial intelligence, engineering design, cognitive science,

The conference focuses on the study and exchange of academic achievements of international researchers. It serves as academic discussion communication platform which is beneficial for research in the field of computer science aiming to stimulate researchers in their work. The collection of papers in this conference proceedings are on Algorithms and Computational Mathematics, Machine learning and optimization, Communication, Networking and Broadcast Technologies and Digital India Initiative.

SMART-2016 received 179 submissions and each submission was reviewed by external reviewers and evaluated on quality, originality, and relevance. The technical program committee accepted 66 papers. As a result, 66 submissions were selected for discussion at the meeting of the program chairs and all 66 papers were accepted for the presentation at the conference. Out of 179 accepted submissions, 66 were chosen to be published in the proceeding of SMART-2016.

The papers presented at SMART-2016 were not its only highlight. In addition to the technical sessions, the conference featured keynote speeches by leading figures from industry and academia, such as 1. Prof. (Dr.) R.K. Mittal, Professor, University School of Management Studies, GGSIP University, New Delhi, and Founder V.C., Teerthanker Mahaveer University, Moradabad, India. 2. Prof. Abu Bakar Abdul Hamid, University Putra Malaysia, Malaysia. 3. Dr. Danila Parygin, Associate Professor, VSTU, Cand. Tech. Sc., Volgograd, Russia. 4. Prof. Raghuraj Singh, Director, Kamla Nehru Institute of Technology, Sultanpur, Uttar Pradesh, India. 5. Dr. Mohamad RidzwanIshak, Sr. Lecturer, Department of Aerospace Engineering, University Putra Malaysia, Malaysia. 6. Prof. Mohammad Sarosh Umar, Professor & Chairman, College of Computer Engineering, Aligarh Muslim University, Aligarh, India. 7. Prof. J. Ramkumar, Associate Professor, Associate Dean for Student Activities (ADSA), Department of Mechanical Engineering and Design Program, IIT-K, Kanpur, India. Chairmain-IEEE UP Section. 8. Prof. (Dr.) Dilip Sharma, Program Coordinator (CSE) & Professor, Department of Computer Engineering & Applications, GLA University, Mathura, India. Secretary-IEEE UP Section. 9. Dr. Aseem Chandel, Professor & Head-EE, BSA College of Engineering & Technology, Mathura, India. 10. Prof. Ekram Khan, Professor, Department of Electronics Engineering, Aligarh Muslim University, Aligarh, India. 11. Prof. (Dr.) Parma Nand Astya, Professor & Dean-CSE, School of Computing Science & Engineering, Galgotias University, Gautam Buddh Nagar, India. 12. Dr. Vibash Yadav, Director, Krishana Institute of Technology, Kanpur, India. In total, there were over 174 individual authors from five countries, representing both academic and industrial organizations. The research areas covered at SMART 2016 were thus not only technical in nature, but reflected the diverse angles from which to approach this emerging research field.

During the organisation of this conference I received help from different people, departments and institutions. First, I would like to extend my sincere thanks to the Chief Patrons Shri Suresh Jain, Chancellor, Teerthanker Mahaveer University, Moradabad and Shri Manish Jain, Group Vice Chairman, Teerthanker Mahaveer University, Moradabad, and the Patrons Prof. R.K. Mudgal, Vice Chancellor, Teerthanker Mahaveer University, Moradabad, Prof. P.D. Kaushik and Dr. Devadesh Sharma, Registrar, Teerthanker Mahaveer University, Moradabad for their valuable support in organising such an international conference. Secondly, I would like to thank the IEEE U.P. Section, India for their enthusiastic assistance and support to this conference. Now, I would like to thank the keynote speakers for their incredible speech on the main theme of the conference. I would like to thank both the authors who submitted their work to SMART-2016 and the external reviewers, who spent their time reviewing the research papers and providing the feedback that resulted in the selection of the papers featured in these proceedings. Several organizations provided financial and logistical assistance in organising the conference, and I would like to acknowledge their support. Lastly, I want to thank to the programme committee members for their diligent and great work that made this conference possible. In conclusion, it was the team effort of all these people that made this conference a success. I welcome suggestions that may help improve the conference in the future. I look forward to this publication providing the foundation for future developments in the field of computer science and information technology.

Prof. Rakesh Kumar Dwivedi
Conference General Chair